

Network Performance Analysis

Experiment date: 2026-09-08 19:25:36 EDT

Input: listed_iperf3_servers.csv

Github Repository Link: <https://github.com/philwisniewski/CS422-Assignment1>

Source public IP: 128.10.2.13 (approx. 40.42260, -86.91070; , United States)

1. Ping Test and Round-Trip Time (RTT)

For each destination that responded to ICMP ping, the script recorded minimum, average, and maximum RTT from the local machine. Public IP geolocation was queried for the source and destination addresses, and great-circle distance was calculated using the Haversine formula.

Destination	Min RTT (ms)	Avg RTT (ms)	Max RTT (ms)	Loss	Latitude	Longitude	Distance (km)
102.214.66.19	193.51	193.76	194.41	0.0%	5.60372	-0.18696	9327.0
102.214.66.39	181.50	181.56	181.69	0.0%	5.60372	-0.18696	9327.0
103.146.200.98	226.85	227.06	227.46	0.0%	-9.43529	147.18000	13693.1
105.235.237.2	237.17	237.33	237.64	0.0%	3.75496	8.78213	10217.1
109.61.86.65	27.12	27.35	27.77	0.0%	N/A	N/A	N/A
109.61.94.65	125.49	125.58	125.77	0.0%	38.72230	-9.13934	6438.0
121.127.43.65	67.35	67.42	67.51	0.0%	N/A	N/A	N/A
128.10.2.13	0.01	0.02	0.03	0.0%	40.42260	-86.91070	0.0
138.199.14.66	153.53	153.66	153.73	0.0%	43.29650	5.36978	7227.7
138.199.4.1	202.55	203.16	203.88	0.0%	N/A	N/A	N/A
138.199.57.129	16.01	16.08	16.17	0.0%	N/A	N/A	N/A
143.244.63.144	186.68	186.88	187.40	0.0%	-33.86880	151.20900	14905.0
148.230.45.213	97.58	99.25	104.16	0.0%	40.41670	-3.70329	6752.7
148.230.59.38	67.92	68.26	69.26	0.0%	37.33870	-121.88500	3026.9
148.230.60.200	130.89	131.43	133.19	0.0%	N/A	N/A	N/A
154.81.51.4	220.90	221.09	221.28	0.0%	-33.86880	151.20900	14905.0
156.146.40.65	123.52	123.62	123.69	0.0%	N/A	N/A	N/A
156.146.52.1	137.13	137.29	137.57	0.0%	41.00820	28.97840	8889.2
156.146.53.53	53.24	53.41	53.73	0.0%	N/A	N/A	N/A
160.242.19.254	209.92	209.97	210.06	0.0%	-8.81466	13.23020	11498.0
169.150.202.193	202.42	202.70	202.99	0.0%	32.08530	34.78180	9972.1
169.150.242.129	124.27	124.35	124.48	0.0%	45.81500	15.98190	7752.7
169.150.252.2	138.46	138.54	138.60	0.0%	37.98380	23.72750	8805.4
173.243.131.29	27.11	27.29	27.52	0.0%	N/A	N/A	N/A
176.23.128.102	207.82	207.99	208.32	0.0%	1.35208	103.82000	15244.1
178.215.228.109	101.61	101.75	101.88	0.0%	50.11090	8.68213	7034.2
185.102.217.170	131.32	131.47	131.65	0.0%	44.42680	26.10250	8454.6
185.102.218.1	148.24	148.52	149.40	0.0%	52.36760	4.90414	6680.6
185.102.219.93	111.45	111.51	111.64	0.0%	50.11090	8.68213	7034.2

Destination	Min RTT (ms)	Avg RTT (ms)	Max RTT (ms)	Loss	Latitude	Longitude	Distance (km)
185.152.65.113	160.01	160.10	160.16	0.0%	50.07550	14.43780	7377.9
185.152.66.67	27.87	27.90	27.93	0.0%	N/A	N/A	N/A
185.152.67.2	54.32	54.54	54.85	0.0%	34.05490	-118.24300	2846.3
185.180.12.40	123.03	123.13	123.28	0.0%	48.20810	16.37130	7614.2
185.246.208.67	120.81	120.88	120.94	0.0%	52.22970	21.01220	7599.3
185.3.160.57	109.67	109.72	109.79	0.0%	50.84760	4.35717	6729.0
185.59.221.51	149.06	149.12	149.17	0.0%	51.50720	-0.12759	6417.9
185.59.223.8	23.40	23.52	23.68	0.0%	40.71280	-74.00600	1089.5
185.76.9.135	123.58	123.78	124.07	0.0%	N/A	N/A	N/A
185.93.1.65	6.00	6.08	6.14	0.0%	N/A	N/A	N/A
185.93.2.193	149.95	150.03	150.12	0.0%	48.85750	2.35138	6710.2
185.93.3.50	114.72	114.76	114.80	0.0%	40.41680	-3.70379	6752.7
189.25.101.151	209.80	210.16	210.85	0.0%	51.50720	-0.12759	6417.9
189.25.97.217	200.39	200.67	201.33	0.0%	25.20480	55.27080	11736.1
189.25.98.63	176.66	176.92	177.60	0.0%	25.03300	121.56500	12163.3
195.181.162.195	31.22	31.25	31.28	0.0%	25.76170	-80.19180	1744.8
197.227.12.18	257.15	257.21	257.24	0.0%	-20.16600	57.50250	15965.4
204.155.243.172	28.92	29.16	29.56	0.0%	N/A	N/A	N/A
207.211.214.65	103.38	103.45	103.55	0.0%	50.82760	4.34389	6729.2
209.40.123.215	44.40	45.01	45.85	0.0%	32.77670	-96.79700	1223.9
212.60.92.134	155.85	155.88	155.97	0.0%	13.44730	-16.66500	7378.5
213.158.175.240	145.42	145.66	145.97	0.0%	30.03830	31.21020	9918.6
217.61.40.96	115.26	115.36	115.71	0.0%	45.69940	9.58110	7358.3
23.249.54.234	34.03	34.37	35.11	0.0%	25.76170	-80.19180	1744.8
23.249.55.42	201.49	201.73	202.27	0.0%	25.20480	55.27080	11736.1
23.249.58.14	186.95	187.56	188.34	0.0%	22.31930	114.16900	12705.1
23.249.60.154	141.32	141.48	141.70	0.0%	35.67640	139.65000	10307.4
37.19.203.1	131.77	131.84	131.88	0.0%	42.69770	23.32190	8422.6
37.19.206.20	24.85	25.14	25.71	0.0%	N/A	N/A	N/A
37.19.216.1	34.34	34.40	34.47	0.0%	29.76040	-95.36980	1411.9
37.19.218.65	126.89	126.94	126.97	0.0%	N/A	N/A	N/A
41.110.39.130	117.16	118.99	120.51	0.0%	36.75980	3.01369	7463.0
41.210.185.162	240.04	240.14	240.35	0.0%	0.31517	32.58160	12430.5
49.205.75.2	216.60	216.88	217.56	0.0%	12.97160	77.59460	13866.5
66.35.20.123	65.72	66.02	66.65	0.0%	37.33940	-121.89500	3027.7
66.35.22.79	16.80	17.16	17.38	0.0%	N/A	N/A	N/A
66.35.27.207	34.59	34.73	34.87	0.0%	32.77670	-96.79700	1223.9
66.35.30.9	47.31	47.67	48.05	0.0%	N/A	N/A	N/A
66.35.31.81	141.41	141.91	142.88	0.0%	35.67640	139.65000	10307.4
69.48.236.198	273.11	273.75	274.20	0.0%	18.52040	73.85670	13165.3
69.48.237.66	135.81	136.79	138.19	0.0%	41.00820	28.97840	8889.2
69.48.238.200	251.63	252.28	254.36	0.0%	25.20480	55.27080	11736.1
69.48.239.124	237.59	238.03	238.72	0.0%	-26.20560	28.03370	13905.8
72.1.86.46	180.92	181.39	182.50	0.0%	-36.85090	174.76400	13176.1

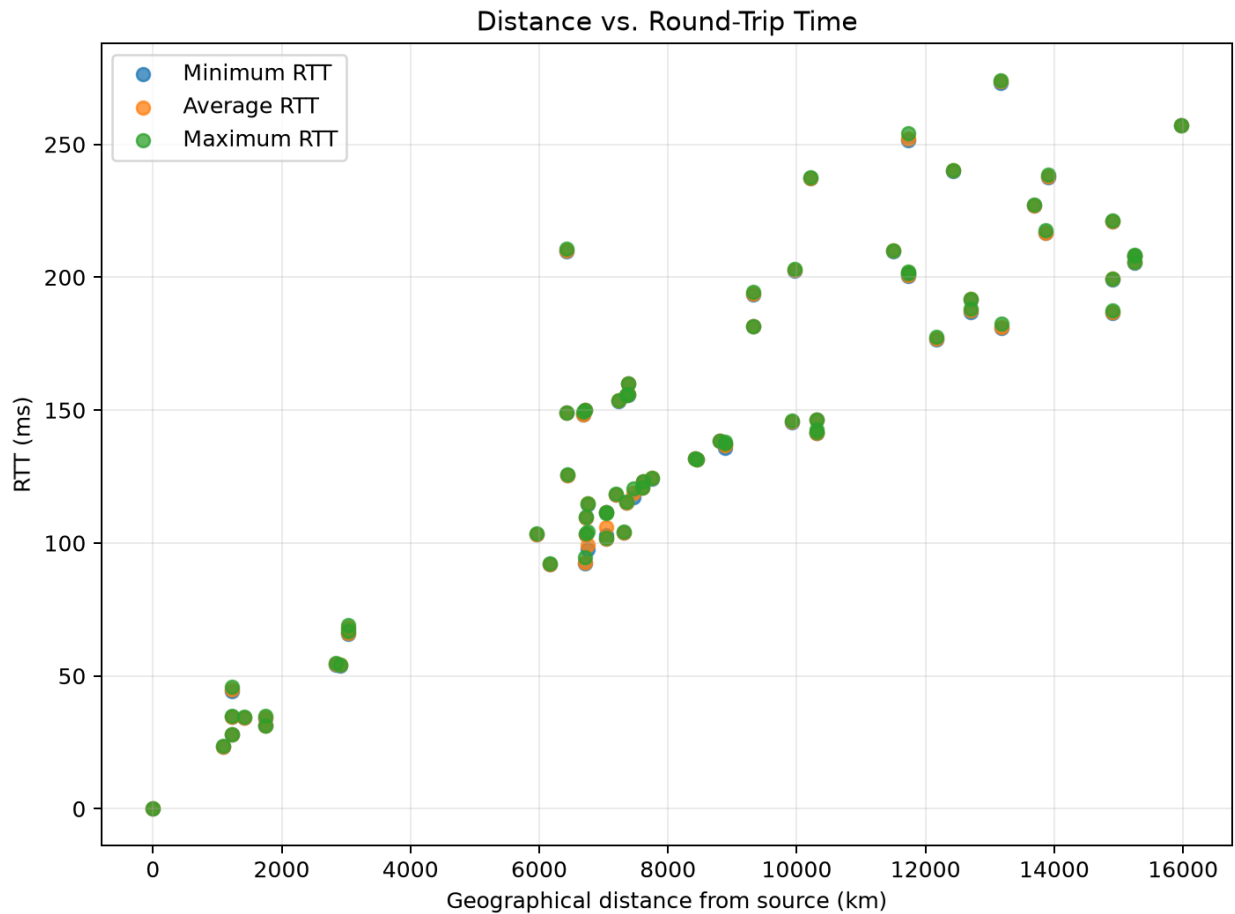
Destination	Min RTT (ms)	Avg RTT (ms)	Max RTT (ms)	Loss	Latitude	Longitude	Distance (km)
79.127.209.1	147.87	148.21	148.71	0.0%	N/A	N/A	N/A
84.17.41.11	54.01	54.11	54.26	0.0%	47.60620	-122.33200	2914.9
84.17.57.129	191.56	191.70	191.85	0.0%	22.31930	114.16900	12705.1
84.17.59.129	155.65	155.74	155.79	0.0%	45.46420	9.18998	7347.8
87.249.137.8	103.31	103.45	103.58	0.0%	53.34980	-6.26031	5955.1
89.187.160.1	146.30	146.35	146.43	0.0%	35.67640	139.65000	10307.4
89.187.162.1	205.56	205.68	205.79	0.0%	1.35208	103.82000	15244.1
89.187.164.1	27.89	27.91	27.92	0.0%	32.77670	-96.79700	1223.9
89.187.165.1	118.05	118.21	118.44	0.0%	47.37690	8.54169	7188.8
96.45.38.22	207.96	208.05	208.21	0.0%	1.35208	103.82000	15244.1
96.45.39.38	102.88	105.86	111.22	0.0%	50.11090	8.68213	7034.2
96.45.40.45	92.02	92.13	92.25	0.0%	53.23940	-2.99056	6161.1
96.45.41.167	103.79	103.95	104.12	0.0%	43.64150	7.00919	7317.2
96.45.42.156	92.30	92.77	94.53	0.0%	48.85660	2.35222	6710.3
96.45.43.6	28.76	28.84	29.00	0.0%	N/A	N/A	N/A
96.45.44.87	199.31	199.41	199.55	0.0%	-33.86880	151.20900	14905.0
a110.speedtest.wobcom.de	106.82	106.92	107.03	0.0%	N/A	N/A	N/A
a204.speedtest.wobcom.de	107.73	107.80	107.86	0.0%	N/A	N/A	N/A
a205.speedtest.wobcom.de	117.01	117.07	117.12	0.0%	N/A	N/A	N/A
a208.speedtest.wobcom.de	113.92	114.02	114.15	0.0%	N/A	N/A	N/A
a209.speedtest.wobcom.de	106.63	106.83	107.38	0.0%	N/A	N/A	N/A
a210.speedtest.wobcom.de	116.90	116.98	117.14	0.0%	N/A	N/A	N/A
a400.speedtest.wobcom.de	109.91	110.03	110.30	0.0%	N/A	N/A	N/A
akl.linetest.nz	179.39	179.44	179.50	0.0%	N/A	N/A	N/A
ams.speedtest.clouvider.net	95.31	95.38	95.44	0.0%	N/A	N/A	N/A
ash.speedtest.clouvider.net	24.69	24.81	24.91	0.0%	N/A	N/A	N/A
atl.speedtest.clouvider.net	13.12	13.40	13.75	0.0%	N/A	N/A	N/A
bwtest.linxtelecom.com	118.42	118.49	118.64	0.0%	N/A	N/A	N/A
chch.linetest.nz	192.44	192.49	192.56	0.0%	N/A	N/A	N/A
chi.speedtest.clouvider.net	6.11	6.24	6.48	0.0%	N/A	N/A	N/A
coverage1.mobicom.mn	199.39	199.44	199.47	0.0%	N/A	N/A	N/A
dal.speedtest.clouvider.net	20.48	20.61	20.68	0.0%	N/A	N/A	N/A
dfw.speedtest.is.cc	58.67	58.81	59.08	0.0%	N/A	N/A	N/A
fra.speedtest.clouvider.net	112.35	112.49	112.85	0.0%	N/A	N/A	N/A

Destination	Min RTT (ms)	Avg RTT (ms)	Max RTT (ms)	Loss	Latitude	Longitude	Distance (km)
gw1.malabo.guineanet.net	236.91	237.14	237.47	0.0%	N/A	N/A	N/A
iperf-ams-nl.eranium.net	95.19	95.30	95.46	0.0%	N/A	N/A	N/A
iperf.as42831.net	88.01	88.11	88.17	0.0%	N/A	N/A	N/A
iperf.fegis.nu	119.71	121.22	127.05	0.0%	N/A	N/A	N/A
iperf.online.net	99.15	99.23	99.33	0.0%	N/A	N/A	N/A
iperf.pendc.com	146.03	146.16	146.32	0.0%	N/A	N/A	N/A
iperf.worldstream.nl	100.15	101.33	105.27	0.0%	N/A	N/A	N/A
iperf1.surfwireless.nl	99.26	99.36	99.42	0.0%	N/A	N/A	N/A
iperf3-vie-at.alwyzon.net	117.34	117.47	117.69	0.0%	N/A	N/A	N/A
iperf3.moji.fr	91.86	91.92	92.00	0.0%	N/A	N/A	N/A
it1.speedtest.aruba.it	126.09	126.44	127.63	0.0%	N/A	N/A	N/A
la.speedtest.clouvider.net	50.77	50.86	51.11	0.0%	N/A	N/A	N/A
lg.ams-nl.gigahost.no	112.79	112.87	112.95	0.0%	N/A	N/A	N/A
lg.gigahost.no	110.55	110.71	110.80	0.0%	N/A	N/A	N/A
lon.speedtest.clouvider.net	88.71	88.81	89.00	0.0%	N/A	N/A	N/A
man.speedtest.clouvider.net	91.18	91.23	91.30	0.0%	N/A	N/A	N/A
nl.speed.vps1.net	96.53	96.62	96.75	0.0%	N/A	N/A	N/A
phx.speedtest.clouvider.net	44.43	44.55	44.61	0.0%	N/A	N/A	N/A
ping-90ms.online.net	190.52	190.62	190.74	0.0%	N/A	N/A	N/A
ping-ams1.online.net	96.15	96.22	96.34	0.0%	N/A	N/A	N/A
ping.online.net	99.14	99.24	99.47	0.0%	N/A	N/A	N/A
sgp.proof.ovh.net	205.09	205.47	206.65	0.0%	N/A	N/A	N/A
spd-icsrv.hostkey.com	125.48	125.55	125.65	0.0%	N/A	N/A	N/A
speed-cb.dimensione.com	125.95	126.16	126.42	0.0%	N/A	N/A	N/A
speed.cosmonova.net	126.07	126.25	126.85	0.0%	N/A	N/A	N/A
speed.mymanga.pro	215.48	218.30	221.20	0.0%	N/A	N/A	N/A
speed.netfiber.net.il	208.86	208.94	209.09	0.0%	N/A	N/A	N/A
speed.rimon.net.il	207.95	208.09	208.29	0.0%	N/A	N/A	N/A
speed1.fiberby.dk	102.82	102.99	103.17	0.0%	N/A	N/A	N/A
speed2.fiberby.dk	102.90	103.11	103.57	0.0%	N/A	N/A	N/A
speedsrv.multinet24.pl	116.70	116.75	116.77	0.0%	N/A	N/A	N/A
speedtest.ams1.nl.leaseweb.net	100.55	100.70	100.91	0.0%	N/A	N/A	N/A
speedtest.ams2.nl.leaseweb.net	100.31	101.64	106.46	0.0%	N/A	N/A	N/A

Destination	Min RTT (ms)	Avg RTT (ms)	Max RTT (ms)	Loss	Latitude	Longitude	Distance (km)
speedtest.chi11.us.leaseweb.net	8.09	8.22	8.60	0.0%	N/A	N/A	N/A
speedtest.dal13.us.leaseweb.net	24.31	24.39	24.50	0.0%	N/A	N/A	N/A
speedtest.fra1.de.leaseweb.net	108.64	108.86	109.07	0.0%	N/A	N/A	N/A
speedtest.goco.ca	25.23	25.29	25.35	0.0%	N/A	N/A	N/A
speedtest.hiper.dk	120.42	120.68	121.10	0.0%	N/A	N/A	N/A
speedtest.hkg12.hk.leaseweb.net	246.59	246.66	246.73	0.0%	N/A	N/A	N/A
speedtest.init7.net	107.71	107.75	107.85	0.0%	N/A	N/A	N/A
speedtest.ip-projects.de	101.34	101.61	102.19	0.0%	N/A	N/A	N/A
speedtest.iway.ch	102.34	102.50	103.08	0.0%	N/A	N/A	N/A
speedtest.kamel.network	110.56	110.75	111.20	0.0%	N/A	N/A	N/A
speedtest.lagoon.nc	233.53	241.79	274.62	0.0%	N/A	N/A	N/A
speedtest.lax12.us.leaseweb.net	50.84	51.23	51.83	0.0%	N/A	N/A	N/A
speedtest.level66.services	105.23	105.31	105.38	0.0%	N/A	N/A	N/A
speedtest.lon1.uk.leaseweb.net	88.00	88.15	88.42	0.0%	N/A	N/A	N/A
speedtest.lon12.uk.leaseweb.net	90.29	90.39	90.59	0.0%	N/A	N/A	N/A
speedtest.masnet.ec	105.34	105.52	105.94	0.0%	N/A	N/A	N/A
speedtest.mia11.us.leaseweb.net	37.72	38.00	38.77	0.0%	N/A	N/A	N/A
speedtest.milkywan.fr	100.38	100.43	100.57	0.0%	N/A	N/A	N/A
speedtest.mtl2.ca.leaseweb.net	26.76	27.16	27.73	0.0%	N/A	N/A	N/A
speedtest.netone.nl	145.69	145.75	145.79	0.0%	N/A	N/A	N/A
speedtest.nl1.mirhosting.net	96.09	96.16	96.18	0.0%	N/A	N/A	N/A
speedtest.nocix.net	16.03	16.10	16.16	0.0%	N/A	N/A	N/A
speedtest.nyc1.us.leaseweb.net	23.75	24.02	24.40	0.0%	N/A	N/A	N/A
speedtest.phx1.us.leaseweb.net	44.83	44.89	44.98	0.0%	N/A	N/A	N/A
speedtest.rapidswitch.com	91.20	91.25	91.30	0.0%	N/A	N/A	N/A
speedtest.sea11.us.leaseweb.net	54.28	54.38	54.52	0.0%	N/A	N/A	N/A
speedtest.sfo12.us.leaseweb.net	52.52	52.55	52.59	0.0%	N/A	N/A	N/A
speedtest.shinternet.ch	103.27	103.40	103.72	0.0%	N/A	N/A	N/A
speedtest.sin1.sg.leaseweb.net	232.99	233.12	233.31	0.0%	N/A	N/A	N/A
speedtest.syd12.au.leaseweb.net	184.57	184.69	184.82	0.0%	N/A	N/A	N/A

Destination	Min RTT (ms)	Avg RTT (ms)	Max RTT (ms)	Loss	Latitude	Longitude	Distance (km)
speedtest.tangerang2.myrepublic.net.id	209.30	209.51	209.67	0.0%	N/A	N/A	N/A
speedtest.tyo11.jp.easeweb.net	154.03	154.19	154.33	0.0%	N/A	N/A	N/A
speedtest.uztelecom.uz	186.31	186.41	186.47	0.0%	N/A	N/A	N/A
speedtest.wdc2.us.easeweb.net	25.00	25.25	25.97	0.0%	N/A	N/A	N/A
speedtest.wobcom.de	106.72	106.74	106.78	0.0%	N/A	N/A	N/A
speedtest.wtnet.de	99.34	99.41	99.49	0.0%	N/A	N/A	N/A
speedtest.xmission.com	34.83	34.92	35.05	0.0%	N/A	N/A	N/A
speedtest1.cwpanama.net	66.97	67.04	67.08	0.0%	N/A	N/A	N/A
speedtest1.flowjamaica.com	47.78	47.93	48.36	0.0%	N/A	N/A	N/A
speedtest1.sox.rs	120.04	120.06	120.08	0.0%	N/A	N/A	N/A
speedtest2.lightningfibre.net.uk	88.08	88.16	88.27	0.0%	N/A	N/A	N/A
speedtest6.cwpanama.net	67.14	67.21	67.29	0.0%	N/A	N/A	N/A
speedtestfl.telecom.mu	257.19	257.25	257.29	0.0%	N/A	N/A	N/A
speedtestlondon.telecom.mu	111.57	111.71	111.86	0.0%	N/A	N/A	N/A
syd.proof.ovh.net	206.04	206.18	206.29	0.0%	N/A	N/A	N/A
t5.cscs.ch	105.25	105.33	105.51	0.0%	N/A	N/A	N/A
yyc-speedtest.xplore.ca	39.59	39.74	39.97	0.0%	N/A	N/A	N/A

Distance vs. RTT



Interpretation: Distance generally places a lower bound on network latency because packets must travel farther which takes longer through the network. However, RTT is not determined by geographic distance alone, as routing paths can be longer than the straight-line path, and congestion and (lack of) infrastructure quality can increase RTT.

The Pearson correlation between geographical distance and average RTT was 0.906, indicating a strong positive linear relationship for this particular measurement set.

Minimum vs. maximum RTT: Minimum RTT is useful as a best-observed latency under relatively favorable conditions. A much larger maximum RTT than minimum RTT indicates increased volatility in RTT time, which may result from increased queueing, congestion, scheduling, or route changes. A large minimum RTT relative to other destinations often indicates a longer or less direct path.

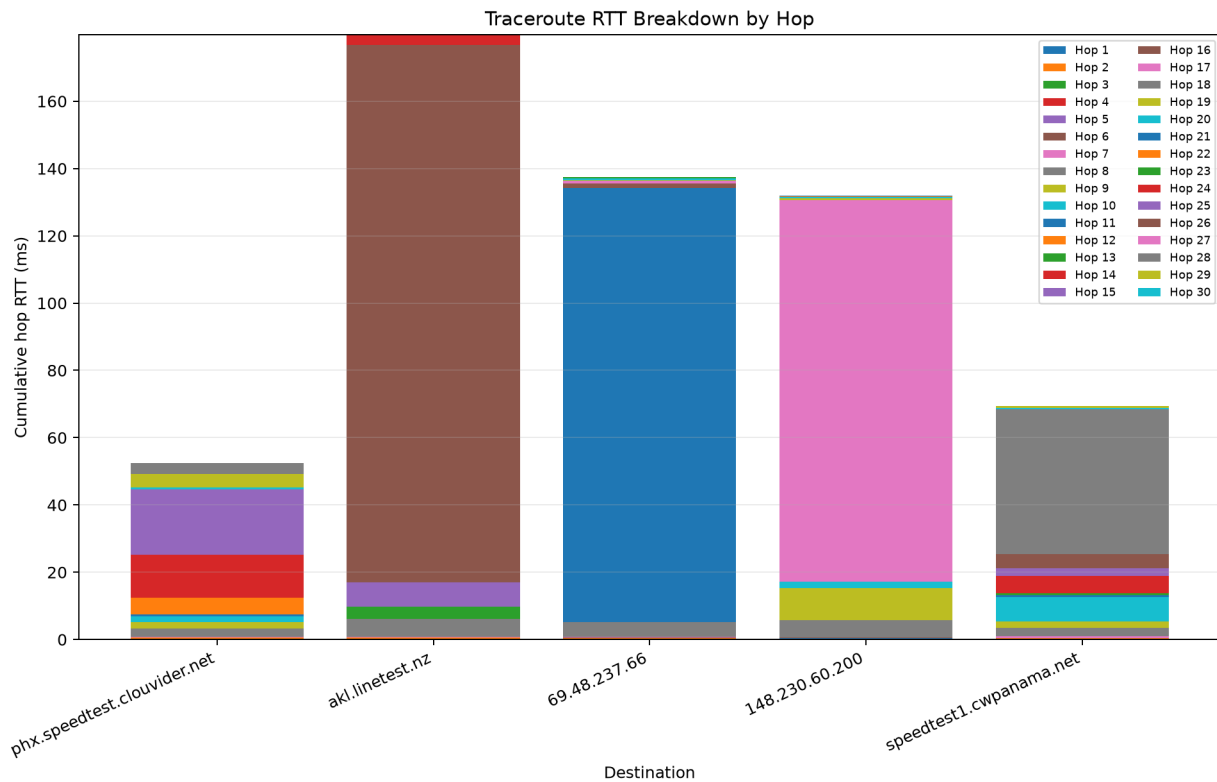
2. Latency Breakdown

Destinations were randomized with a fixed seed, then tested until up to five destinations with successful end-to-end traceroutes were found. Traceroute used three probes per hop. Non-responsive intermediate hops are retained as missing values; they are not assigned invented latency. A destination is included in the analysis only when traceroute clearly reaches it.

Destination	Final hop	Responsive hops	Destination RTT (ms)
phx.speedtest.clouvider.net	20	19	44.60
akl.linetest.nz	24	23	179.70
69.48.237.66	20	19	137.01
148.230.60.200	22	19	131.51

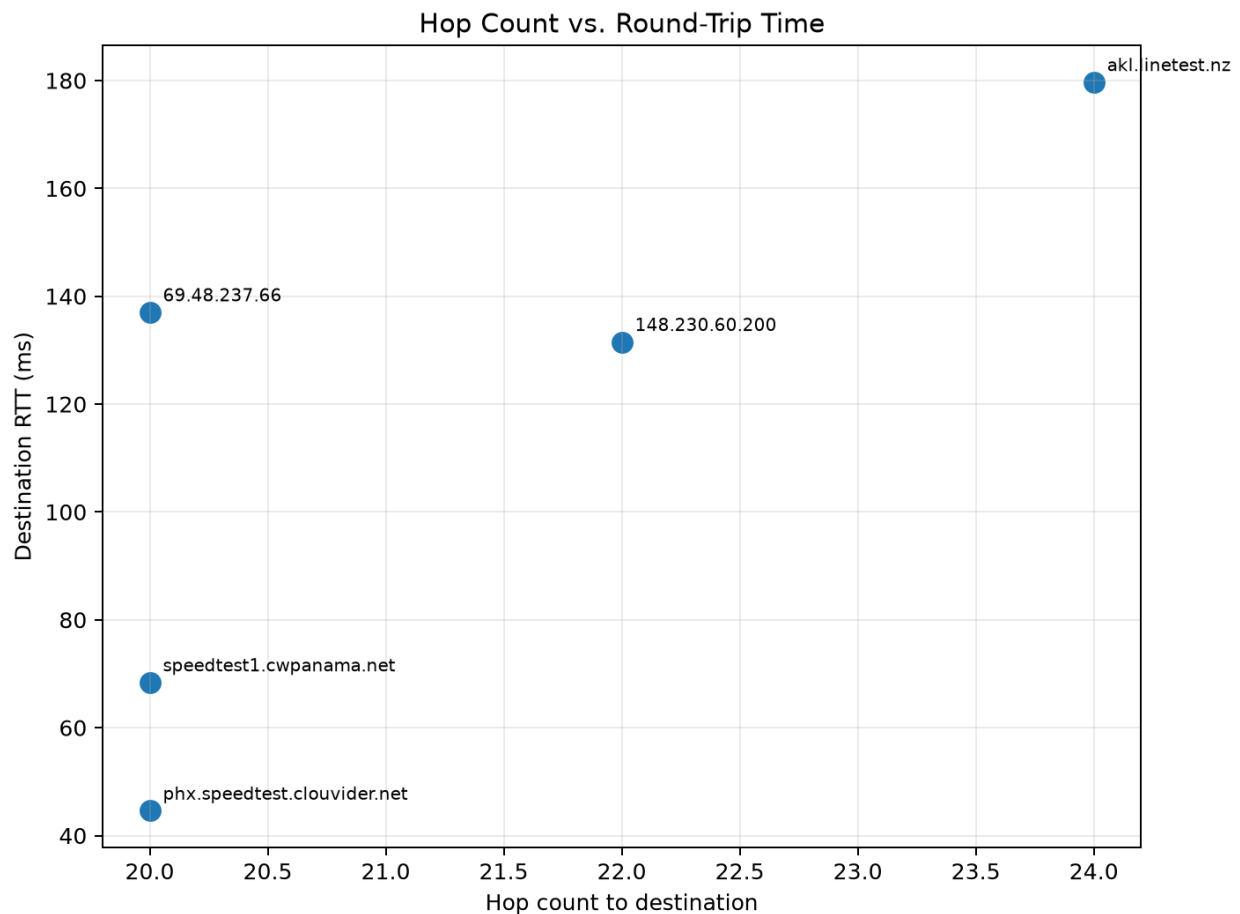
Destination	Final hop	Responsive hops	Destination RTT (ms)
speedtest1.cwpanama.net	20	18	68.40

Traceroute Latency Breakdown



Plot note: calculates RTT deltas by subtracting previous hop RTT from current hop RTT. Non-responsive hops are left empty rather than inventing latency values.

Hop Count vs. RTT



Interpretation: More hops often correspond to higher RTT because each additional router introduces propagation, transmission, processing, and queueing opportunities. However, this is not always the case, as some hops are much closer to each other than others and can take less time.

The Pearson correlation between hop count and destination RTT was 0.786. With only five destinations, this is descriptive rather than statistically conclusive.

Corner cases: A '*' in traceroute means that a probe did not receive a response from that hop. This does not necessarily mean packets cannot pass through that router. The script skips destinations that fail to reach the final destination so that the hop-count/RTT analysis does not treat incomplete paths as successful measurements.

Reproducibility and Code Map

This report was generated by the same one-shot Python script used to run the measurements and create the figures. The relevant code sections are:

Task	Functions / section
Input / server discovery	load_servers_from_csv(), load_servers_from_web()
Public IP + geolocation	get_public_ip(), geolocate()
Ping / RTT	ping_one()
Distance calculation	haversine_km()
Traceroute	traceroute_one()
Distance-vs-RTT plot	make_distance_rtt_plot()

Task	Functions / section
Stacked latency plot	make_stacked_bar_plot()
Hop-count-vs-RTT plot	make_hop_scatter_plot()
PDF report	NetworkReport, build_report()
One-shot orchestration	main()